

Stakeholder Groups

Farmers

Who they are: Smallholder farmers across Nigeria's agro-ecological zones who submit natural language queries describing crop symptoms, pest damage, or soil conditions, spanning routine agronomic questions to urgent issues like disease outbreaks or livestock-threatening conditions.

Why they are important to the project: They are the direct end users of the retrieval outcome; the entire purpose of the project, faster and more accurate access to agronomic knowledge, is measured by whether it actually reaches them with information they can act on. A retrieval system that scores well on benchmarks but fails farmers in practice (e.g., surfacing irrelevant documents for urgent disease queries) would undermine the project's core goal.

What matters most to them: Getting relevant guidance quickly, especially when a crop issue threatens their harvest or livelihood. They also want the system to understand their own words and local terminology, not just formal agronomic vocabulary, and confidence that the information is trustworthy enough to act on without needing to verify it elsewhere.

Agricultural Extension Officers

Who they are: The field staff who currently provide in-person agronomic advice, and who will use the retrieval system's output (or work alongside it) as a supplement to their own guidance once deployed.

Why they are important to the project: They are the operational users whose adoption determines whether the tool succeeds in practice. Their direct field exposure to farmer questions gives them the clearest view of where retrieval succeeds or fails, and their willingness to trust and recommend the tool directly affects whether it actually reduces the extension services bottleneck.

What matters most to them: A tool that extends their reach without replacing their judgment, especially in ambiguous or high-stakes cases (e.g., a symptom that could indicate a fast-spreading disease). They want retrieved documents they can vouch for, an easy way to flag poor results, and confidence that the system supports rather than undermines their role as trusted local advisors.

Potential Benefits and Risks

Farmers

Benefits: Faster access to relevant agronomic knowledge, less dependence on extension officer availability, and higher-stakes issues (e.g., a spreading crop disease) surfacing accurate documents sooner.

Harms/concerns: Misretrieval of urgent queries (e.g., a disease symptom returning generic soil-health documents) causing real crop or income loss; unequal retrieval quality for farmers who describe symptoms in dialect or non-standard phrasing; loss of trust if irrelevant or incorrect

guidance goes unflagged.

Ethical, social, cultural, and economic considerations:

- **Ethical:** fairness in retrieval quality — performance should not degrade for farmers based on regional dialect, literacy level, or how they phrase a query.
- **Social:** risk of eroding trust in digital agronomic tools if farmers feel their queries return unhelpful or irrelevant results.
- **Cultural:** the knowledge base's coverage may not fully represent regional farming practices or local names for crops, pests, and diseases across Nigeria's diverse agro-ecological zones.
- **Economic:** delayed or incorrect guidance on disease or pest issues carries direct financial risk through crop loss.

Extension Officers

Benefits: Less time spent manually searching reference material, more time for hands-on farmer support, and a tool that surfaces edge cases (ambiguous or low-confidence queries) for their review rather than silently guessing.

Harms/concerns: Over-reliance on retrieved results leading to reduced scrutiny of the system's accuracy; erosion of independent diagnostic expertise over time; concern that the tool signals reduced need for their role rather than support for it.

Ethical, social, cultural, and economic considerations:

- **Ethical:** responsibility for acting on retrieved guidance in high-stakes cases should remain shared between the system and the officer, not silently deferred to the tool.
- **Social:** officer buy-in and professional standing could be affected if the tool is rolled out without their input on what "good" retrieval looks like in the field.
- **Economic:** reduced search time could be framed as freeing capacity for more farmer visits, or negatively as justification for reduced staffing, depending on how deployment is positioned.

Engagement Method

Farmers — Distributed Dialogue

Why the method is appropriate: Farmers are a large, dispersed population with brief, one-off contact with the system at the moment they submit a query. A lightweight, distributed, asynchronous method fits this reality far better than a workshop or standing committee.

Why it is suitable for this group: It respects that most farmers want an answer, not a seat on a panel, while still gathering signal on whether farmers whose queries use non-standard or dialect phrasing are being underserved.

How the engagement would be conducted: A short, optional feedback prompt would appear after a query returns results, asking whether the farmer found the documents relevant and useful, alongside a low-friction way to flag results that missed the mark. Responses would be aggregated periodically (e.g., monthly) to identify patterns across many farmers rather than reviewing individual queries case-by-case.

Extension Officers — User Committee

Why the method is appropriate: Extension officers are a bounded, identifiable group who can be convened repeatedly over the life of the project, unlike farmers who interact with the system only briefly.

Why it is suitable for this group: Officers have sustained, hands-on exposure to farmer queries and local conditions, making them uniquely positioned to spot recurring retrieval failures. A committee gives them real, ongoing influence over how the system evolves rather than having it imposed on them.

How the engagement would be conducted: A small standing committee of extension officers (rotating membership across regions to capture different agro-ecological zones) would meet regularly (e.g., biweekly or monthly) to review a sample of recent queries and retrieved documents, flag systematic gaps in the knowledge base, and discuss proposed changes to ranking or coverage before deployment. This formalizes expert review into a structured, accountable process with real decision input.

Influence on Project Design

Dataset/Knowledge base: Farmer distributed dialogue would surface query phrasing styles or regional terms underrepresented in the current knowledge base, informing targeted document collection. Extension officer feedback would identify systematically poor-performing query types (e.g., certain pest categories), guiding where additional curated documents are most needed.

Retrieval design: Recurring committee review could inform decisions such as adjusting ranking thresholds for high-stakes query types (e.g., requiring higher relevance confidence before surfacing disease-related documents) or flagging low-confidence retrievals for mandatory officer review rather than direct farmer display.

Outputs: Feedback could shape how results are presented, for example, showing a confidence indicator or a short set of ranked options rather than a single forced top result, so officer or farmer judgment remains meaningfully part of the decision.

Evaluation: Both groups' feedback would push evaluation beyond aggregate nDCG@5 toward per-region and per-query-type performance, ensuring the system is not judged "successful" while systematically underperforming for certain dialects, crops, or urgent categories.

Deployment decisions: Concerns from both groups support a staged rollout: initially running the system in an assistive capacity (surfacing documents for extension officers to vet) rather than

direct-to-farmer deployment for high-stakes categories, with broader farmer-facing access expanded only as trust and evidence of reliability accumulate.